

The Brick Pavilion

Refaat Nassif
Architect

From Design to Construction Process The Construction of a Dome Pavilion

Latakya, Syria
2021



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P R O F I L E

This Project began as personal interest while ending my scholarship at Ecole Speciale d’Architecture in Paris, France. It consisted in challenging new Designing process. Assembling Local artisan skills, with high technology tools (CNC) and CAD Software. Taking into consideration local skills in the very early stage of the design, making the design distinct and very handy to third parties: Customer, designer, and artisans.

E D U C A T I O N

June 2009

Ecole Speciale d’Architecture of Paris
Master 1 in Architecture

c o n t a c t

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ANTIQUITY ARCHITECTURE DOMES

EARTH DOME, SYRIA
Dates back to (7000 BC)

Since the first civilizations domes are present...





ANTIQUITY ARCHITECTURE DOMES

DAMASCUS CITADEL, Syria
Built in 1076

ANTIQUITY ARCHITECTURE DOMES

Sayyida Ruqayah Shrine Syria



gettyimages
Luis Dafos



The Brick Pavilion
Architect Refaat Nassif

**ANTIQUITY
ARCHITECTURE DOMES**

King's College Chapel
England Built in 1441

—1 INTRODUCTION:

In the Medieval period, the unity between Architectural and civil engineering, had reflected on methods of building. Impressed by its accentuated geometry combining aesthetic and structural advantage for these edifices.

One can notice that Gothic (قوطية) in western Europe and Islamic Architecture in general from Spain to India during the Omayyad caliphate, the famous ornamented vaulting style as known as Muqarnas are very good examples of the proper utilization of the applied science (Math, Physics..) and its obvious influence on the previously mentioned architectural styles.

Domes were the answer when covering vast areas was demanded, The stability of the structure relied on a high level of knowledge of geometry (Graphic Statics). In accordance, and in order to understand and launch this project we had to go through analyzing and observing existing dome shaped monuments in addition to reproducing plans; as there were no original plans in ancient archives.

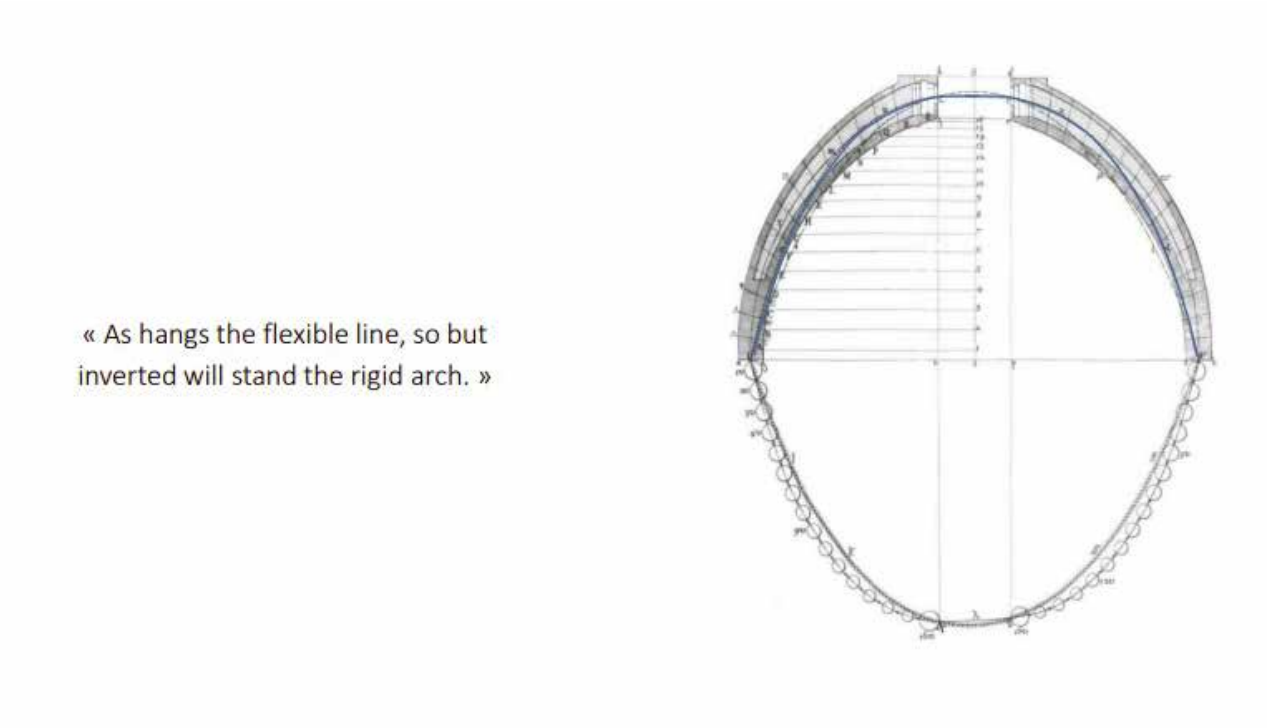
The evolution of the arch began with the roman style, considered to be the first arch used in construction. Geometrically it is the half of a circle. Then it evolved through time changing its curvature to offer a lighter and taller structure like gothic arches close to a parabolic curve.



2 Material:

Structural Geometry

The arch becomes a structural and architectural element.

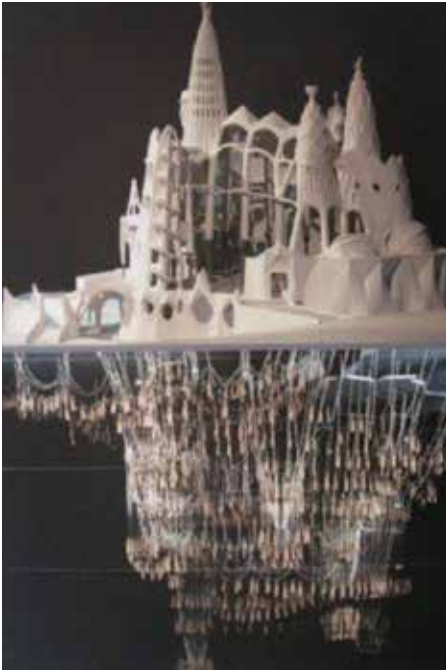


Hanging chain.



Old builders were master builders that knew how to observe nature and get inspired from it, to develop new construction ideas. The geometric stability is obtained through the shape: like the eggshell. Very resistant, the proportion between the thickness of the shell and the span of the egg shows clearly the important role of the shape.

In the Mid century, master builders relied a lot on stone material. From this material they mastered techniques to adapt it to their constructions, and how it behaves under compression forces. That technique is named Graphic statics. Gaudi used it for his experiments and his buildings like the sagrada familia.



—3 PROCESS & TOOLS:
-Block construction and concrete pouring

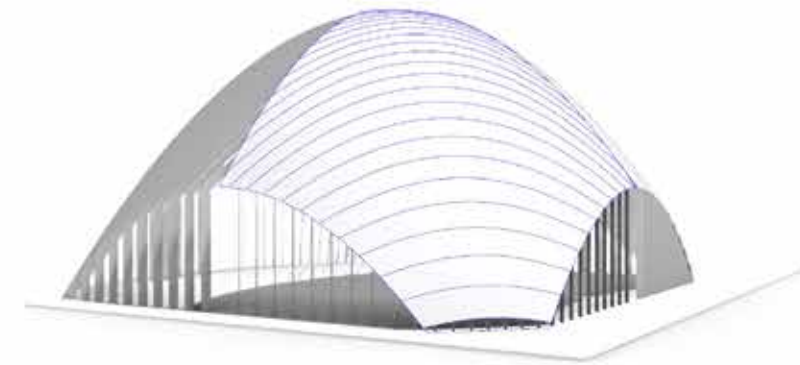
First of all, we had to simulate the figure using 3D programs in order to understand its limits and its capability for division, the main reason for this was that we have been studying the possibility of applying this method on the models; At the same time, we had to calculate every example to achieve the right geometry that could be constructed using the chosen materials at any scale.

Consequently, we tried to find the best mesh geometry considering the chosen material by breaking the Dome down into smaller geometry. This figure will provide the necessary information for the designer to control the construction process, and economize effort and material for the construction. The main approach in design was to dismantle the whole Dome into pieces which could be assembled (built) in a construction site, we used the same strategy to tackle the pitfalls, we divided them into much smaller steps. For example, about choosing the perfect material: although we built 3 mini models in wood, bricks, and reinforced concrete, we chose the “bricks” model over the rest.

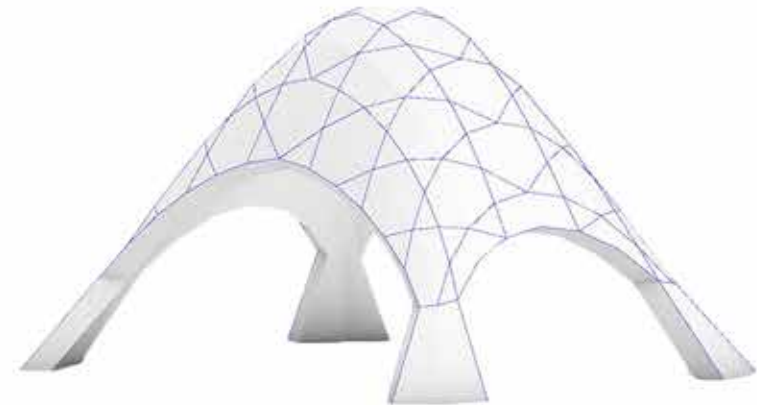
- 1- Choosing the right Geometry
- 2- Accurate as much as possible to the design.
- 3- A rigid formwork serving as support for the construction.
- 4- A formwork with less material possible.

This part of the study essentially focuses on how to build the physical model from the 3d design model. From here came two strategies : block assembly and strips, each has a different process in modeling and construction. For experimenting mold’s techniques and pouring, in this study the block assembly part overtook the strips part.

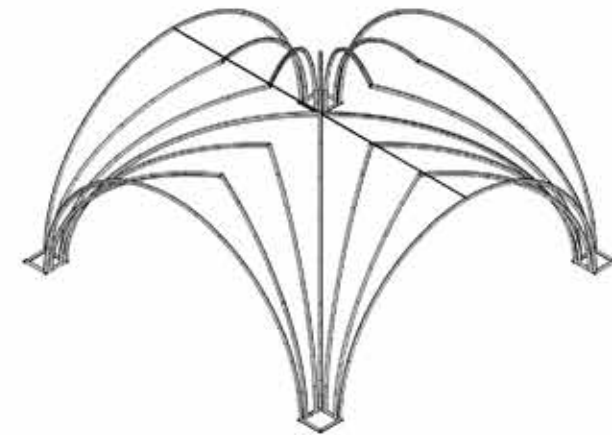
Forms	Strips		Block Assembly	
	1		Arch	
	2		Triangulation	
	3		Polygon	



Strips



Polygons



Arches



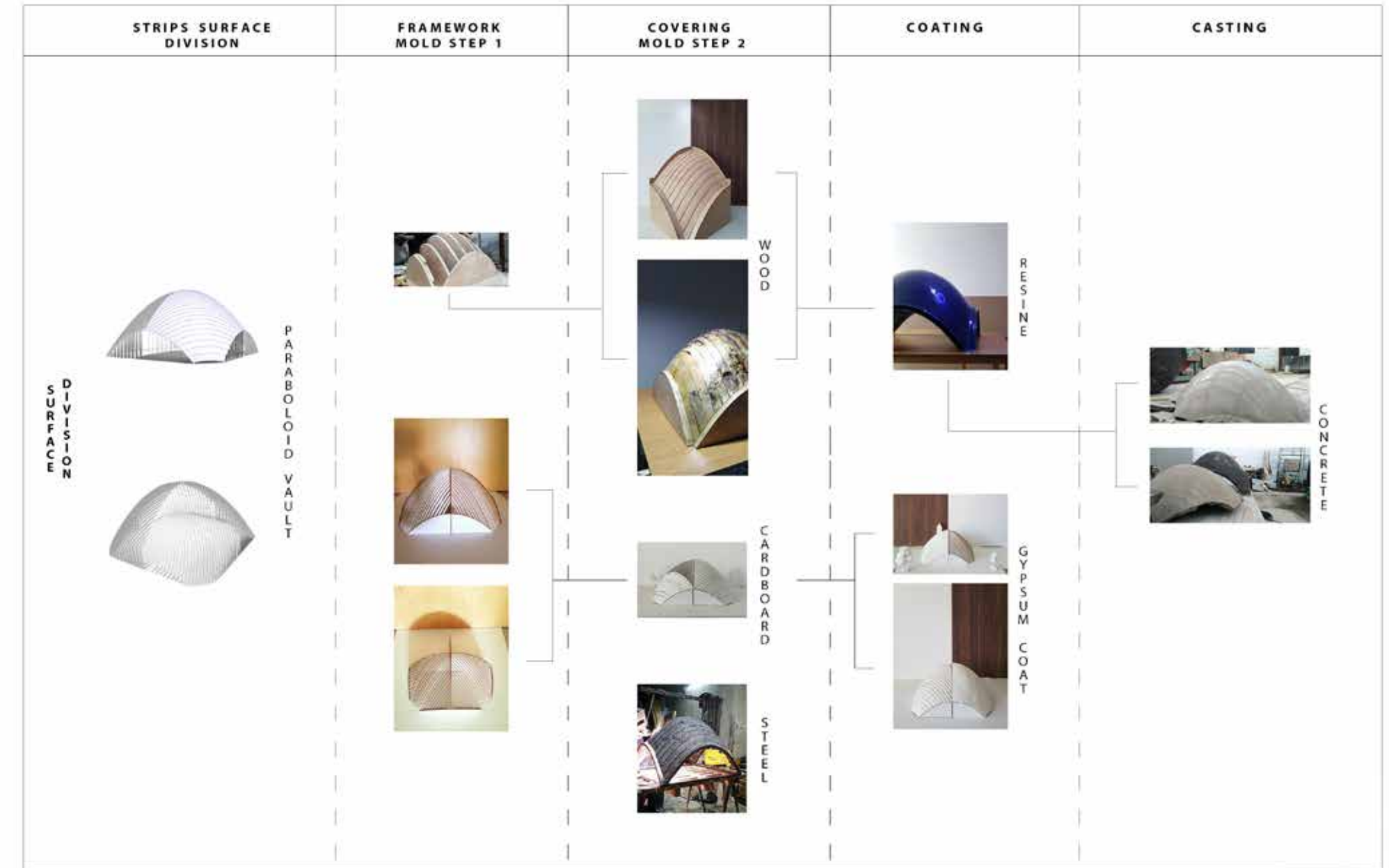
Triangulation

A-Strip coverage (Bending Wood):

We simulated a figure also that would take into consideration covering for a concrete shell. Thanks to the 3d program Rhinoceros and Grasshopper, we achieve this by drawing strips over the dome surface. Using wood as coverage material, to represent the dome curvature, and to bend the wood, we used thin strips of 1cm thickness. So the thin layer of bended wood is the main formwork of the dome on the top and will hold the concrete shell, while It reaches its maximal strength. As said before, bending wood needs thin material thickness, and this forces us to superpose these strips in multiple layers to have the required security and the stability of the formwork. Which implies more wood material, more work, and more cost.

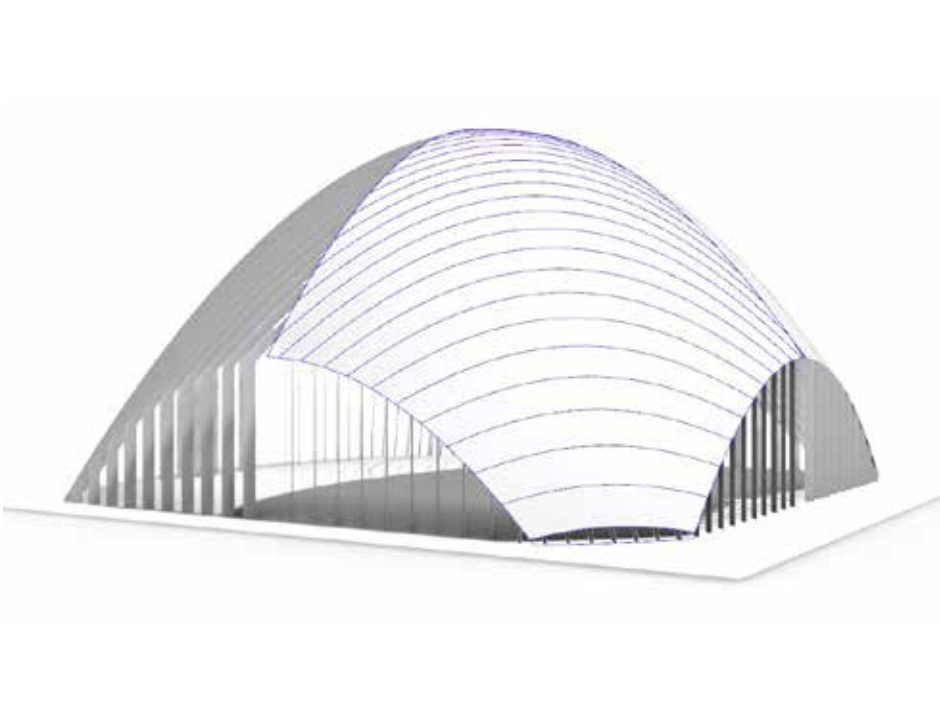


- Process (Strips covering)



A-Strip coverage [Process]:

This method consists of four steps:



1- Design the shape and divide the surface with Rhinoceros.



2- Prepare the first element of the formwork: The beams.



3- Covering the beams with bended strips.



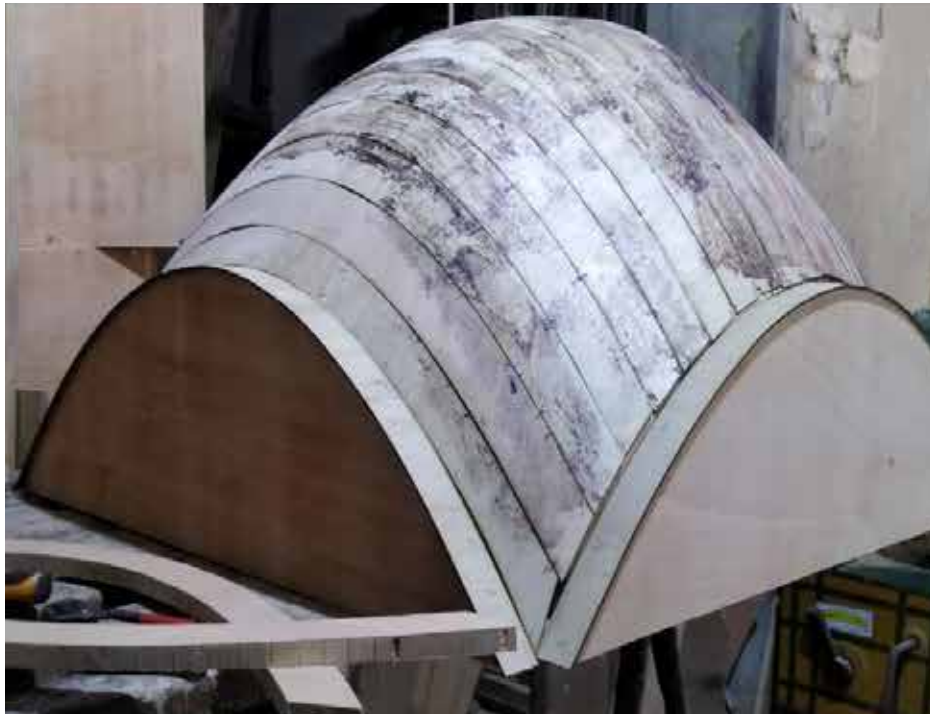
4- Coating the surface of the dome.

A-Strip coverage (Mold step 1):

This model, a dome, a simulation of the construction constraints with concrete material. The covering represents a real challenge because of its curved shape. After the making of the model, resin coat and fiberglass comes to cover the wood mold. As a result a new fiberglass mold will be made, then concrete can be casted later. This mold is a fiberglass and resin mold, the reason why using this material is because it can be used 8 to 10 times. It has excellent physical and mechanical properties and low shrinkage.



First Element of the shape, Plywood framework.



Covering the framework with strips.

A-Strip coverage (Coating Fiberglass & Resin):

After finishing the wood mold, resin and fiberglass are applied to form the second mold.



Fiberglass and resin mold.

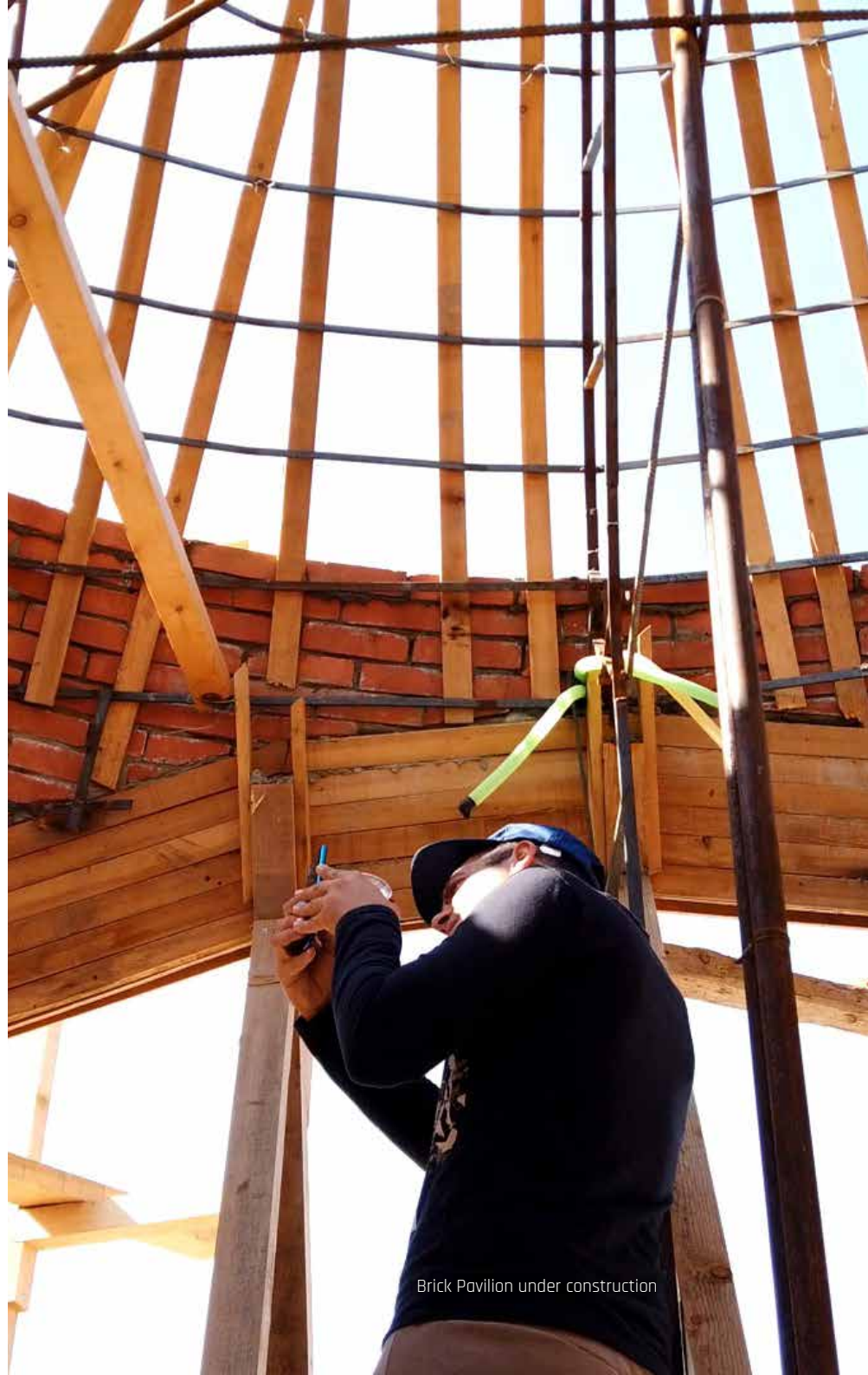
A-Strip coverage (Concrete casting):

In case of coating the mold, only one mold is needed while in case of pouring two molds are needed: one positive and one negative. We choose pouring which is a fully experimental process to explore the capacity of the mold and the casting. Secondly we used concrete without any ferrocement. The demolding, due to pressure inside the mold produced by the heat. What we retained from this experiment is that a very smooth surface was obtained thanks to the mold (positive and negative) that wouldn't have been possible by coating. The concrete dome at a scale of one meter diameter supported itself without any rebars or any other steel material (ferrocement). For the mold, the absence of backbone didn't help for demolding which led to some extremity parts to break.



B-Blocks Assembly :

We generated every piece with 3d printing, then used this piece to form silicone molds, then we plaster casted the pieces. We used wood sticks as trusses to support the construction and then the building would hold itself as we went on elevating without any support. And finally, we had assembled the Model. Afterwards, we had to move to a bigger scale and although this might seem easy, it was the opposite; as we tried to apply the calculated designs, we started to face various pitfalls, such as the perfect material, the perfect mesh, the perfect proportions and finally the most economical solution of all the mentioned above. For example one of the difficulties, was 3d printing and casting in real life would have cost a lot more...But would be very possible,thanks to the multiple studies, designing, fabricating molds, and finally assembling (arches Guidelines, Triangulation, Polygons).



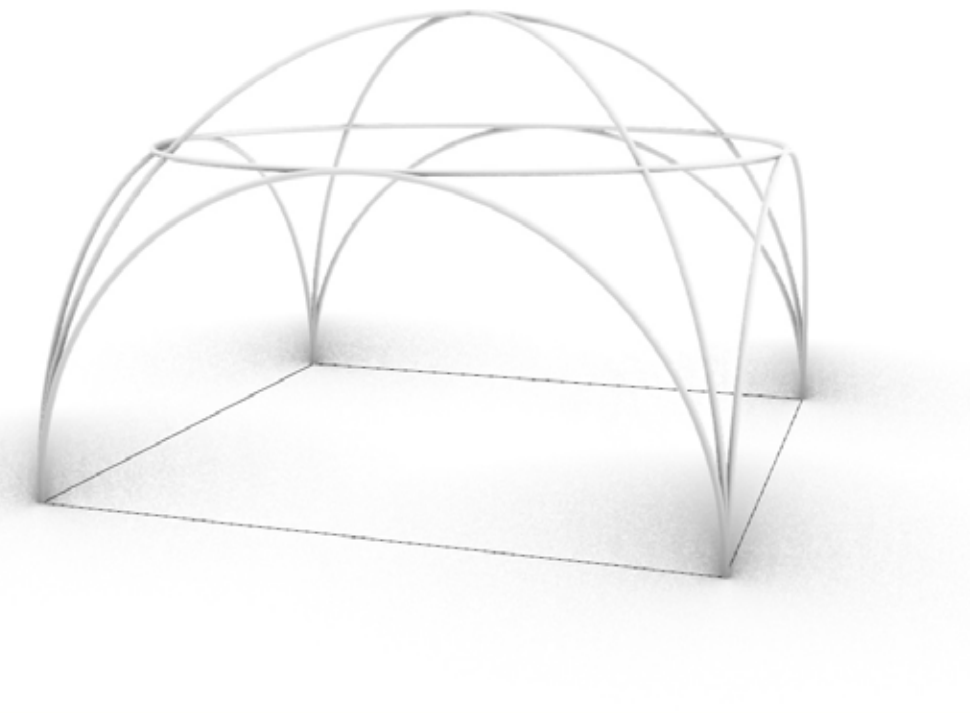
Brick Pavilion under construction

- Process (Arches as Guidelines)

ARCHES GUIDELINES	CASTED BLOCKS MOLD STEP 1	PLACING ARCHES	ASSEMBLY
CROSS VAULT FAN VAULT V A U L T T Y P E	GYPSUM BLOCKS	SQUARE SEGMENTS DIAGONAL SEGMENTS	CROSS VAULT FAN VAULT

B-Blocks Assembly (Arches as Guidelines):

Cross vaults are defined by two different arches. One is repeated on the square edges. The other for the diagonals. These arches are half circle curves.



Cross Vault Arches (3d drawing).



Metallic wire (helper, guideline).



Cross Vault (Gypsum model).

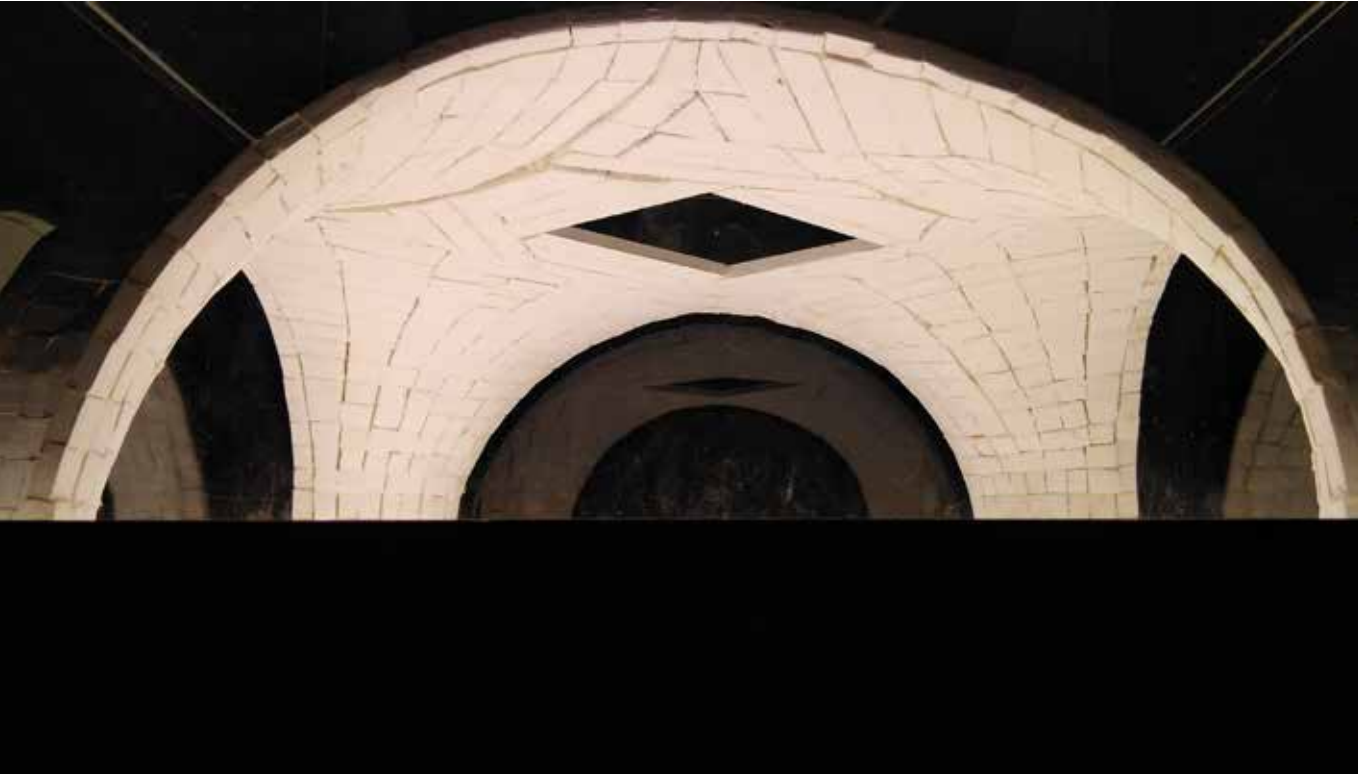


Cross Vault, Gypsum model 25cm*25.

B-Blocks Assembly (Arches as Guidelines):



Cross Vault, Gypsum model 25cm*25.

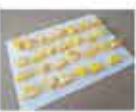


Fan Vault, Gypsum model 50cm*50.



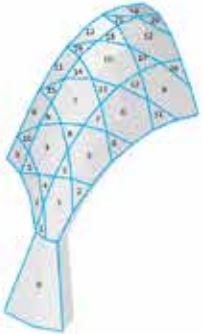
Fan Vault, Gypsum model 50cm*50

- Process (Triangulation)

BLOCKS MESH DIVISION	3D PRINTED BLOCKS MOLD STEP 1	SILICONE CASTING MOLD STEP 2	GYPSUM CASTING MOLD STEP 3	ASSEMBLY
MESH BLOCKS DESIGN     CONFIGURATIONS	     BLOCKS ASSEMBLY BLOCKS	 MOLD  SILICONE CASTING	 GYPSUM BLOCKS 	 ASSEMBLY  DOVE



- Process (Polygons)

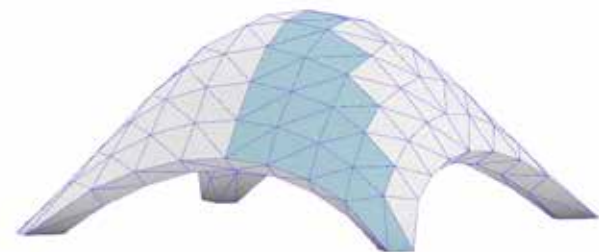
BLOCKS MESH DIVISION	3D PRINTED BLOCKS MOLD STEP 1	SILICONE CASTING MOLD STEP 2	GYPSUM CASTING MOLD STEP 3	ASSEMBLY
MESH BLOCKS DESIGN  CONFIGURATIONS	 3D BLOCKS PRINTED	 MOLD CASTING SILICONE	 GYPSUM BLOCKS	 DOME  ASSEMBLY



B-Blocks Assembly :

Triangulation and Polygons

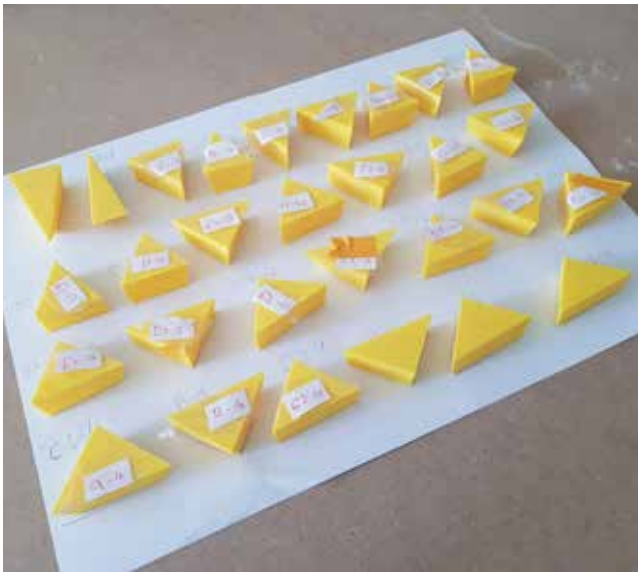
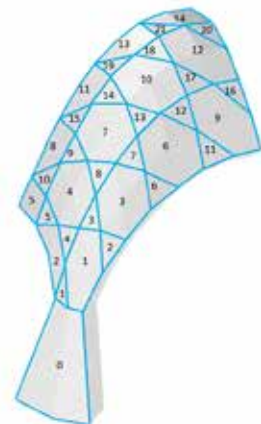
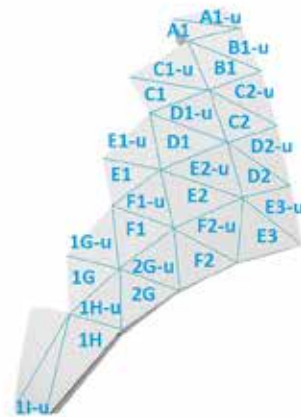
Design the shape by identifying the patterns.



Triangulation pattern and blocks identification



Polygons pattern and blocks identification



3d printed blocks.



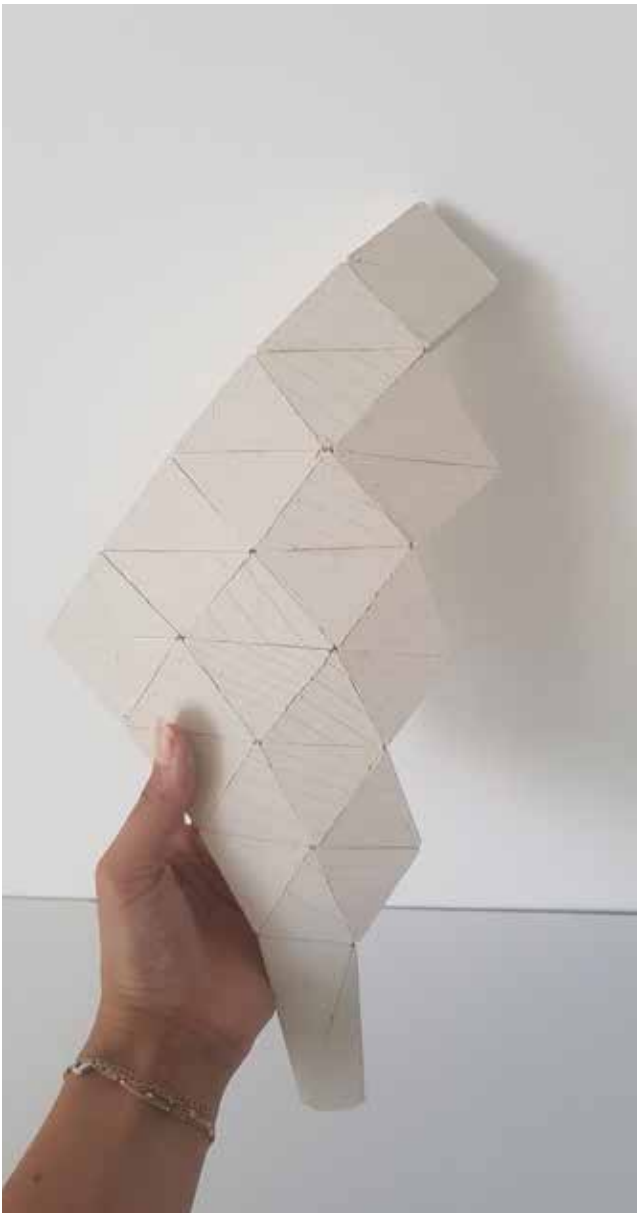
Silicone molds.



Gypsum blocks.



Gypsum Casted Blocks.



Blocks Assembly.

B-Blocks Assembly (Conclusion):

We tried to build upon every step we took, we utilized computers in building and analyzing our models to become a model, we took advantage of the model in using wood pillars and trusses to build our 1X1 meter model; Hence, two main ideas were starting to take shape, one is to use a permanent wood support structure (Method 2), the other one is to use a temporary one, which helps during first phases of building and then we could take it to another construction site (Method 1).



4 THE PROJECT :

Vault as a model

The main Theme was a traditional one. Since the beginning of civilization domes have been present. With that in mind we tried to think and design authentic homes in today's environment, where the rural landscape is filled up by concrete construction, optimizing the profit. What makes this practice usual. While the environmental question is the new concern. In our case tradition is somehow the answer to this irregularity between. The trial of this project is to combine material and design in an efficient way. For that we imagined original houses defined by its shape and its relationship with the material used. At the same time, brick or concrete can be used, so that the shape is used to orient the forces to only compression forces. This is very important in the strategy: the shape is totally aware of the material that will be used for its Structural behavior. The pavilion was designed like an eggshell, inspired by its unique shape and strength. Designed with Rhinoceros, Grasshopper, and Kangaroo Plugins. The Pavilion shape is the result of the Program membrane shell simulation or sometimes also called hanging net. A Dome shape on a rectangular base. The curvature of the dome is closer to the Catenary Curve or hanging chain curve. When the purpose is residential then one can build multiple domes and connect them in series to have a fully functional house with as many rooms as needed.



A-The Design :



A-The Design :



B-THE Site :

Our target is rural areas, thus we built the first model in Latakya's countryside, in a village called Debash, which is almost 35 kilometers from Latakya and 500m above sea level. The site was offered by my family to encourage this project. It was set on a village house terrace.



B-THE Site :



C-METHOD 1 :

Off-Site building elements (method 1)

1-Wooden Arch Frames:

Two sets of concrete arch formwork were constructed to define the boundaries of the dome. Framework were constructed off-site, then brought to the location.

2-Measurements:

35 Reference point's coordinates were taken from the design program. Slab supports were added to respect the dome envelope during the construction. They also helped to draw the curvature of the dome, each point having its coordinates. Their heights differ to specify the different reference points.

3-Coverage :

Wooden strips were bended to shape the dome. This aims to test the characteristics of the wood. Covering the top surface of the dome, as a formwork for the concrete shell. With multiple layers of bended wood before the final step: installing rebars and pouring concrete of the dome shell.



C-METHOD 1 (Workshop work) :



Arches Frame built in workshop The wood work was done in a carpentry shop and transported to the construction site.



Wooden Arches Frames. Frameworks constructed off-site.



Arches Formwork transported to site.

Concrete arch formwork

Two sets of wood trusses were shaped as an arch-frame, each one has its dimensions (two big arches and two small ones). Those four arches would define the dome boundaries, as its main function is to hold the building (support). The other function of this wooden arch framework is to form the concrete arch formwork.



C-METHOD 1 :

35 Reference point's coordinates were taken from the design program, as landmarks. Represented by Slab support. Drawing the dome envelope.



35 Reference point's coordinates, their heights differ to specify the different reference points.



C-METHOD 1 :

Additional support was added to support arches formwork from one part and to define the shape's envelope from another part.



C-METHOD 1 (Wooden strips) :

Wooden strips were bended to shape the dome. This aims to test the characteristics of the wood. Covering the top surface of the dome, as a formwork for the concrete shell. With multiple layers of bended wood before the final step: installing rebars and pouring concrete of the dome shell.



C-METHOD 1 (Wooden strips) :



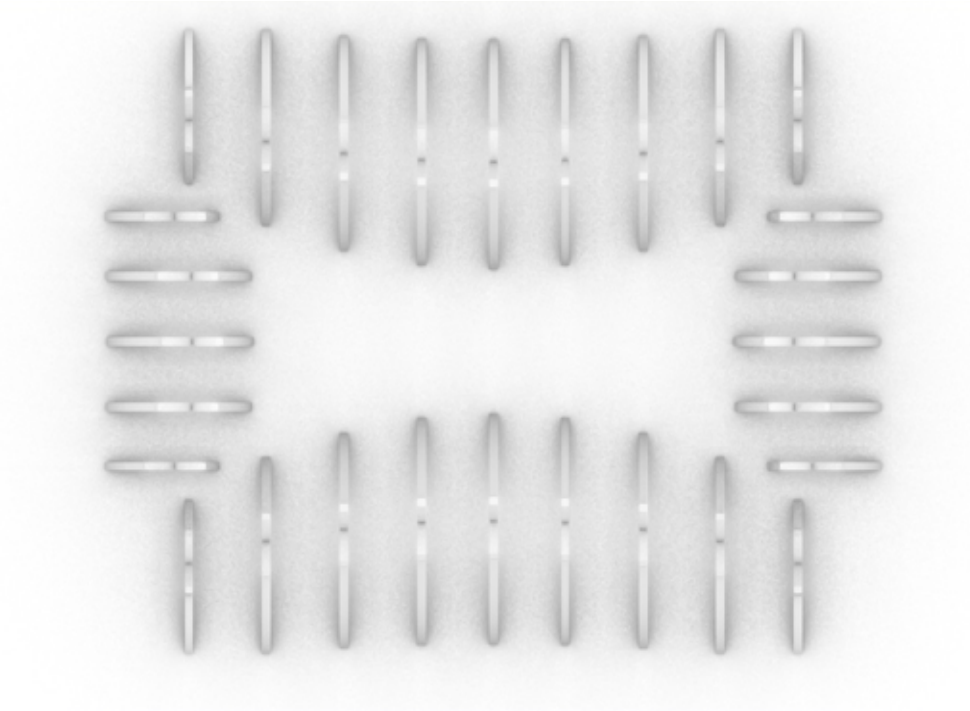
D-METHOD 2 (Final dome) :

This method was totally inspired by the 3D model, it consists of having different heights of supporting pillars that would shape the dome curvature. Those pillars hold the curved beam that will transfer the loads while building from the dome to the ground through the pillars, and the number of pillars used covers the need of calculating its structural properties. In addition to this, there is a steel frame that has two purposes; one is to be a guideline as its ring-shaped horizontal lines indicate where the bricks should go, the other one is to prevent bricks from falling during construction. The formwork is cut in small pieces to shape the curve and it's fixed onto the steel frame. In this case the curve would almost be identical to the 3D design and every brick will be built in its right position, this method allows the designer to build it entirely with concrete or bricks or even both.

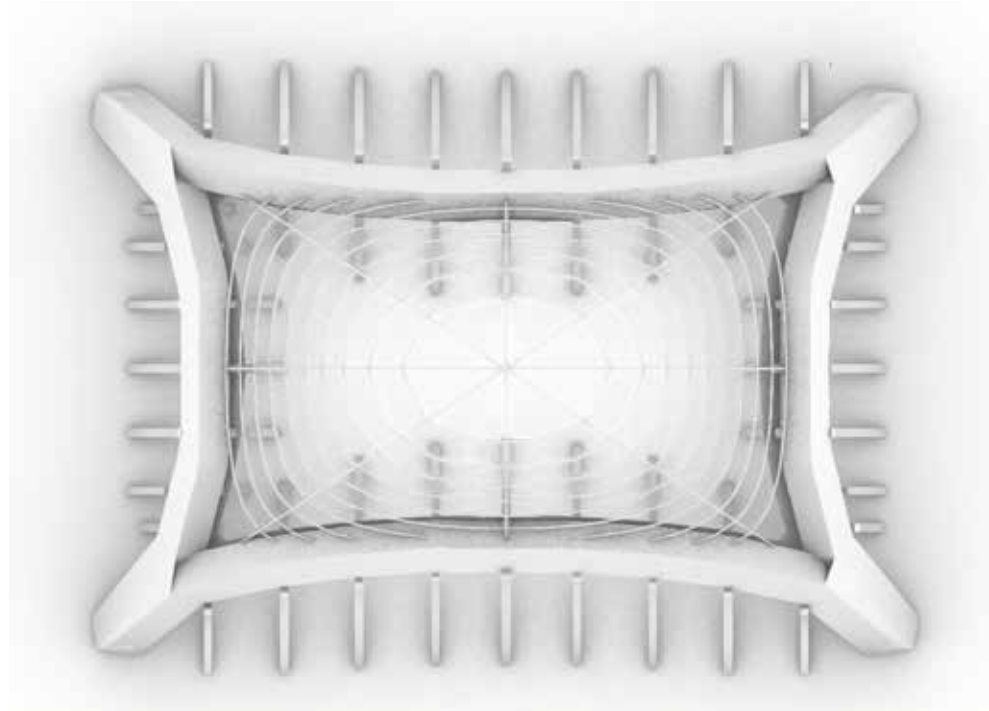


D-METHOD 2 :

The 3d model helped to explain each step of the construction process.



Pillars plan view.
Used to support the construction



Pillars plan view.
Concrete arches formwork



Pillars forming the arch curves , and the support.

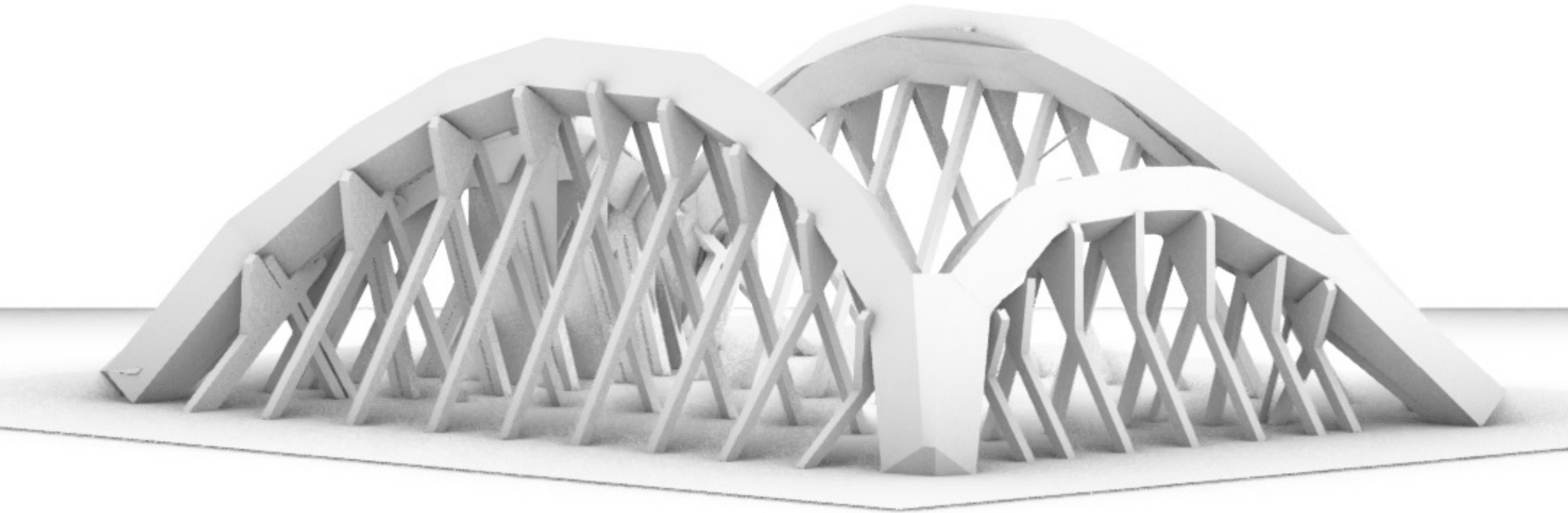
D-METHOD 2 :



Pillars.



Two sets of concrete arch formwork.



Pillars drawing and supporting the concrete arch formwork.

D-METHOD 2 :



Concrete arches formwork.



Concrete arches formwork

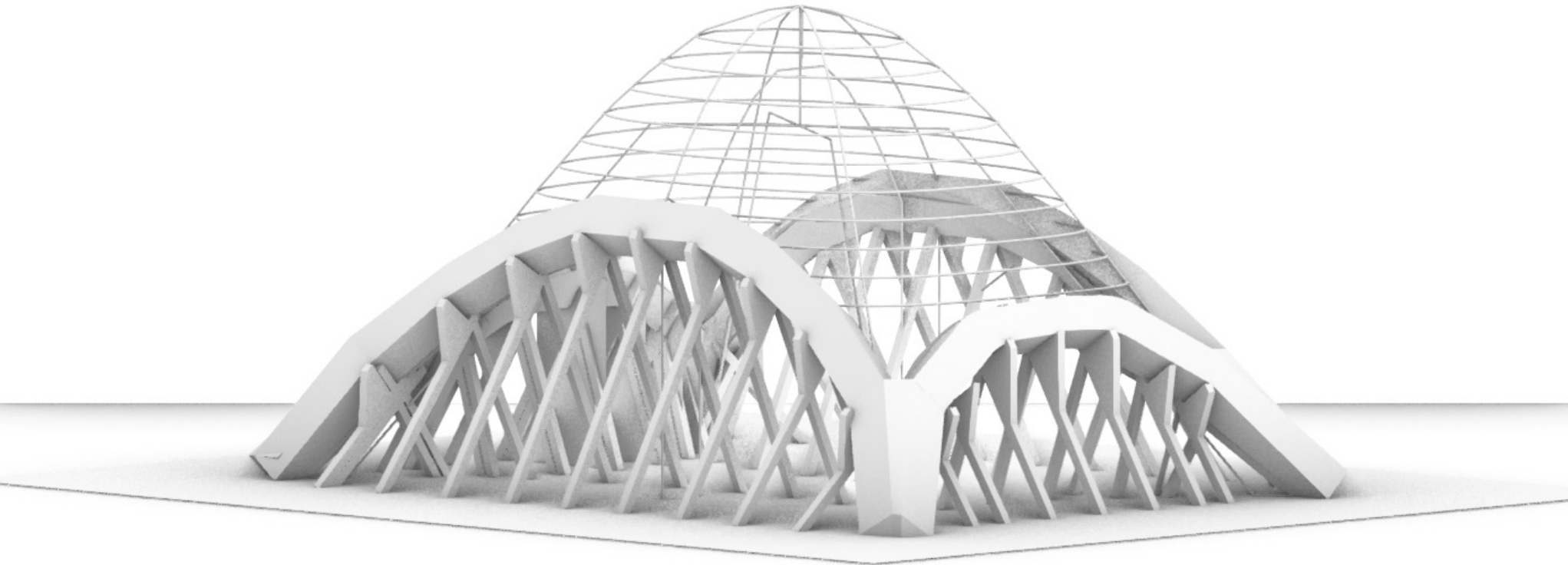


Concrete arches formwork.



Concrete arches formwork.

D-METHOD 2 (Arches and Guidelines) :



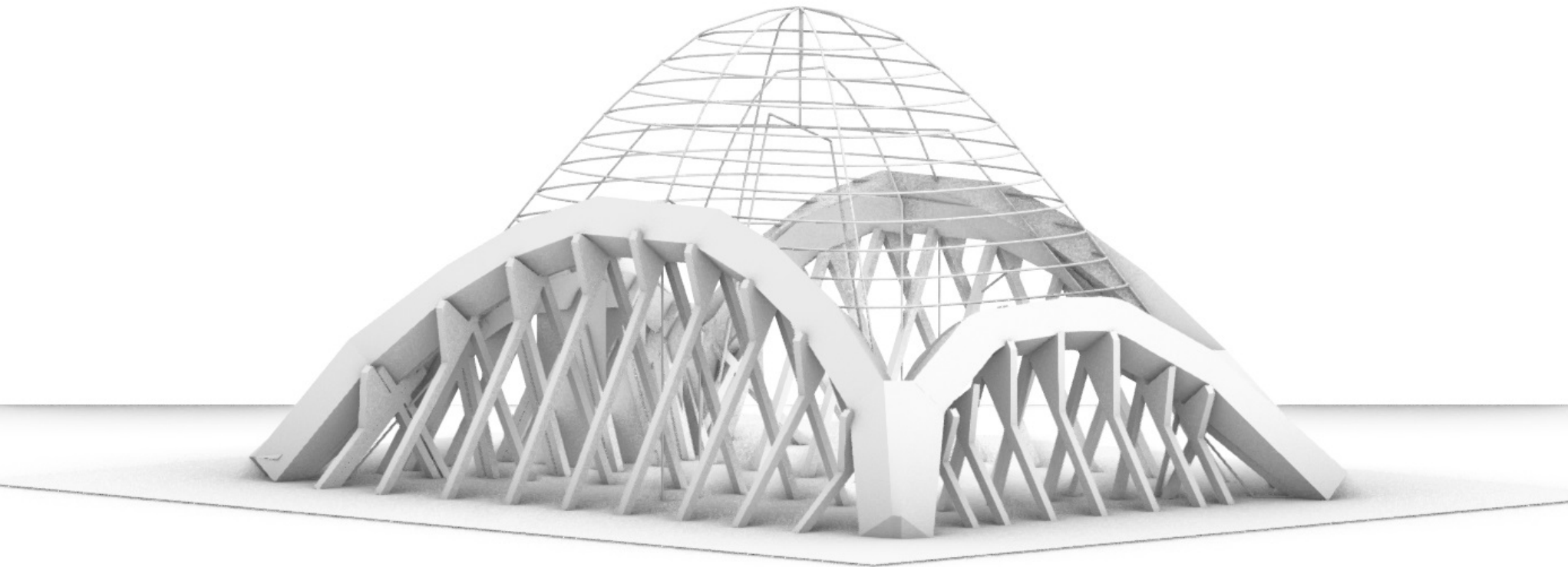
Arches formwork.







D-METHOD 2 (Ring-shaped guideline) :



Ring-shaped guideline.



Ring-shaped guideline.





Brick construction.



Brick construction.



Brick construction.



Brick construction.

Brick construction



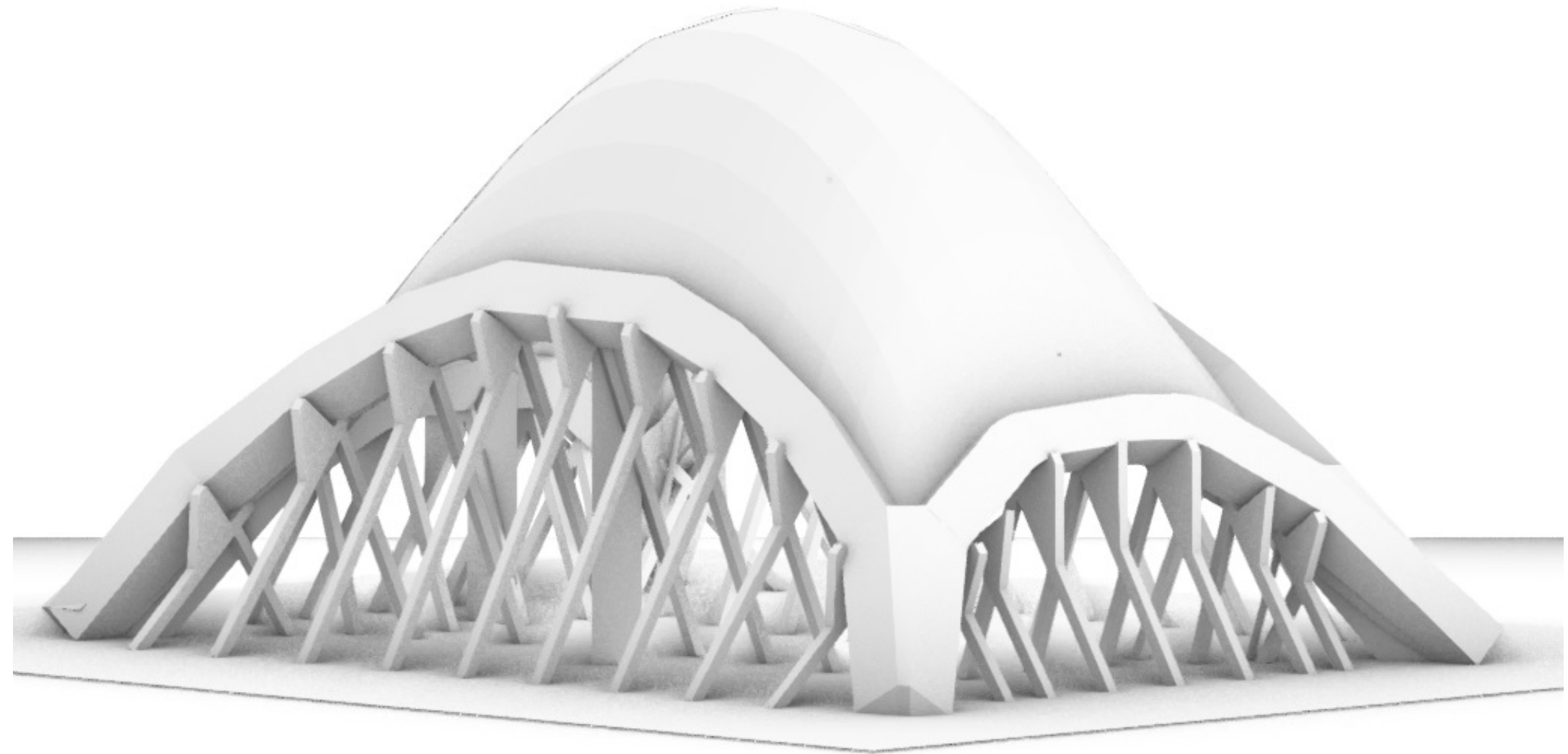
5 CONCLUSION :

In both methods, we used concrete for the foundations and the principal beams, the arches. Representing the ribs of the dome. Then the bricks started. We used 50*50 square individual foundation which has been calculated previously to withstand the weight above and distribute it to the ground.

The main difference is that in method A one gets a formwork that can support the construction anywhere but it would require more numbers of workforce, while in method 2, it requires a limited number of workers and it can be built on site. Another difference is that in method 1 the framework is independent from the supporting pillars where in method 2 the pillars are part of the framework. Not to forget that in the bending wood in method 1 is relatively expensive, on the contrary there is no need for bending wood in method 2, we used instead small pieces of wood that were put in the same way of the 3d model; In other words, in method 1 we could not imitate the 3d model precisely because of wood bending properties, in contrast, using small pieces of wood in method 2 allowed us to control the curve of the dome easily. In addition to this, bending wood should have multiple layers to withstand the weight of concrete without deforming.

To conclude, the first method was not completed due to lack of time and the high costs of building two domes, and this experiment was interrupted by method 2, which was much easier in the aspects of:

- 1) work force (it required fewer numbers compared to method).
- 2) total costs, as the first method requires a premade formwork and bended strips of wood.
- 3) time, for method 2 was faster than method 1.



The Brick Pavilion

Refaat Nassif
Architect



Final dome



Final dome.



Final dome.



Final dome.

The Brick Pavilion

Refaat Nassif
Architect







Thanks to :

- [@My Parents](#), for offering the site.
- [@My wife and my son](#), for the support.
- [@Professor Philippe block & John Ochsendorf](#), for their remarkable work which was very inspiring.
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- [@Kinana Mohamad \(Architect\)](#) exterior design and image render.
- [@Mohamad Abdallah \(Architect & Teacher\)](#) exterior design and image render.
- [@Mohamad Salim \(Architect\)](#) exterior design and image render.
- [@Maher Salloum workshop](#) (cnc milling machine) mold’s patterns cutting.
- [@Asheer shanata workshop](#) (cnc milling machine).
- [@Louay Taraf and samir Taraf](#) mason builder.
- [@Ali Salameh](#) formworker.
- [@Firas Sabe al lel](#) carpenter.

Reference :

- [Christiaan Huygens and the Problem of the Hanging Chain](#)
John Bukowski
- [The combination of tile vaults with reinforcement and concrete](#)
David López López, Tom Van Mele & Philippe Block
- [Equilibrium systems Studies in Masonry Structure](#)
Philippe Block
- [Structural Design in the Work of Gaudi](#)
Santiago Huerta
- Funicular analysis.
- Hanging chains.
- Hanging nets.
- Parabola.
- Catenary.
- Stability rather than material strength.
- Distributed load.
- Compression forces only.
- Force density.
- Form Diagram- Force Diagram.
- Analysis tools.
- Rhinoceros CAD Software.
- Grasshopper Plugin.
- Kangaroo Plugin.

Keywords :

- Architectural heritage.
- Historic structures.
- Research.
- Brick vault.
- Curved masonry structures.
- Masonry shells.
- Masonry domes.
- Thrust Network approach.
- Network Analysis.
- Equilibrium.
- Graphic statics.
- Geometry controlled loads.
- Geometric rules.
- Structural Geometry.

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